RECOMMENDATION

UnitedHealth Group (UNH): Long

Base-case DCF value $526.64 per share against an as-of price of $367.08: +43.5%.

$354.47	$526.64	$764.68
bear (-3.4%)	base (+43.5%)	bull (+108.3%)

As of 2026-09-30. The value is an intrinsic-value estimate, not a dated price target. The bear case sits *below* the price, and the result depends on a low beta (slide 11). Educational project, not investment advice.

WHY

Thesis in three bullets

1. **The price assumes little growth.** The reverse DCF says the price implies `6.0%` stage-1 FCF growth. UNH's median rolling 5-year FCF growth is `13.7%`.
2. **Revenue kept growing.** Revenue grew from `$287.6B` (FY2021) to `$447.6B` (FY2025). The problem is margin, not demand.
3. **The thesis is a margin recovery, and it is not yet in the numbers.** Operating margin fell from `8.7%` to `4.2%`. The long case needs FCF to stop shrinking.

FUNDAMENTALS

Business snapshot

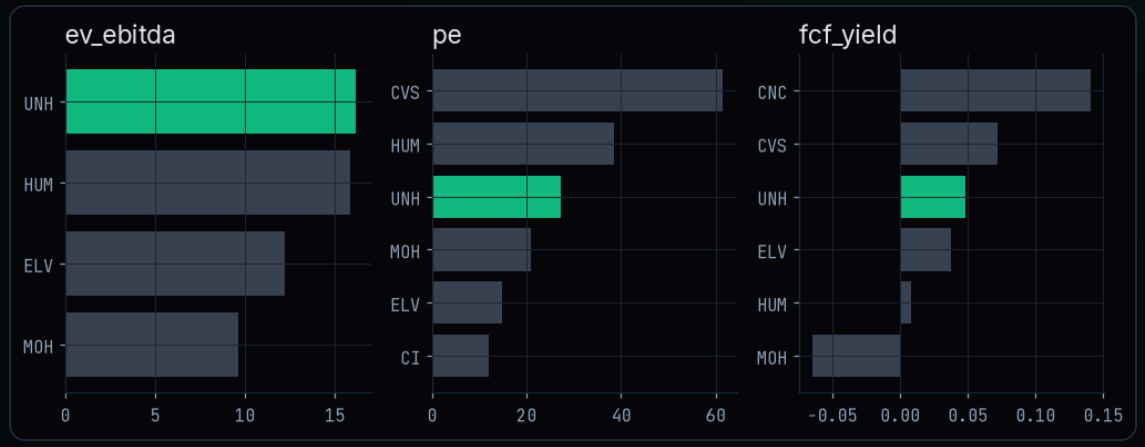
Fiscal year	Revenue	Operating income	Op. margin	FCF	FCF margin
FY2021	$287.6B	$24.0B	8.3%	$19.9B	6.9%
FY2022	$324.2B	$28.4B	8.8%	$23.4B	7.2%
FY2023	$371.6B	$32.4B	8.7%	$25.7B	6.9%
FY2024	$400.3B	$32.3B	8.1%	$20.7B	5.2%
FY2025	$447.6B	$19.0B	4.2%	$16.1B	3.6%

SEC EDGAR 10-K figures as first reported. FCF = operating cash flow − capex. Revenue keeps rising while operating income and FCF shrink. That gap is the key driver to watch.

RELATIVE VALUE

Comps: a premium on earnings, a discount on cash

	EV/EBITDA	P/E	FCF yield	Gaps
UNH	16.2x	27.3x	4.9%	
ELV	12.2x	14.9x	3.8%	
CI	n/a	12.1x	n/a	capex not reported in XBRL; total debt not reported in XBRL
HUM	15.9x	38.6x	0.8%	
CNC	n/a	n/a	14.1%	net loss: P/E not meaningful; negative EBITDA: EV/EBITDA not meaningful
MOH	9.6x	20.9x	-6.4%	
CVS	n/a	61.5x	7.2%	total debt not reported in XBRL
Peer median	12.2x	20.9x	3.8%	

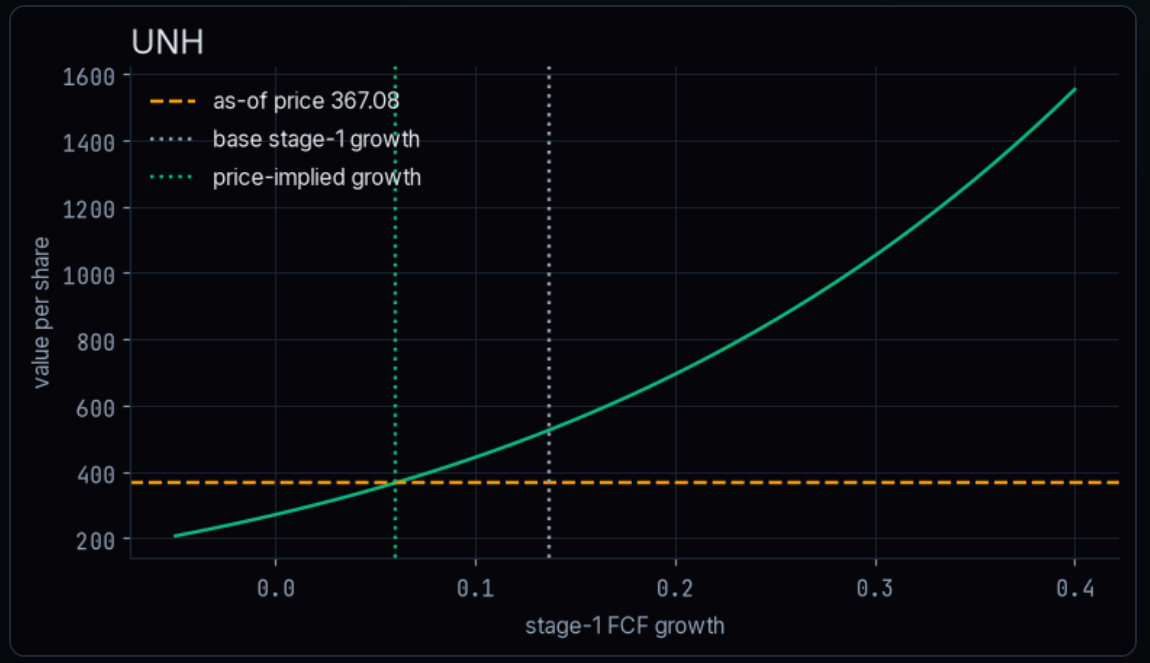


UNH trades above the peer median on P/E and EV/EBITDA but offers a higher FCF yield. Comps alone don't make the case. Gaps are left blank, never estimated.

VALUATION

DCF and reverse DCF: what the price assumes

risk free	5.3%	FRED DGS10 on 2026-09-30
beta	0.41	5y daily stock vs SPY returns to as-of (OLS slope)
equity risk premium	8.3%	Ken French Mkt-RF, arithmetic mean 1926-07 to 202608, x12
cost of equity	8.7%	CAPM: rf + beta x ERP
pre tax cost of debt	5.5%	FY2025 interest expense / average total debt (SEC 10-K)
tax rate	12.9%	FY2025 income tax / pretax income (SEC 10-K)
equity weight	82.0%	market cap (as-of price x Yahoo shares) / (market cap + book debt)
wacc	8.0%	we x ke + (1 - we) x kd x (1 - t)

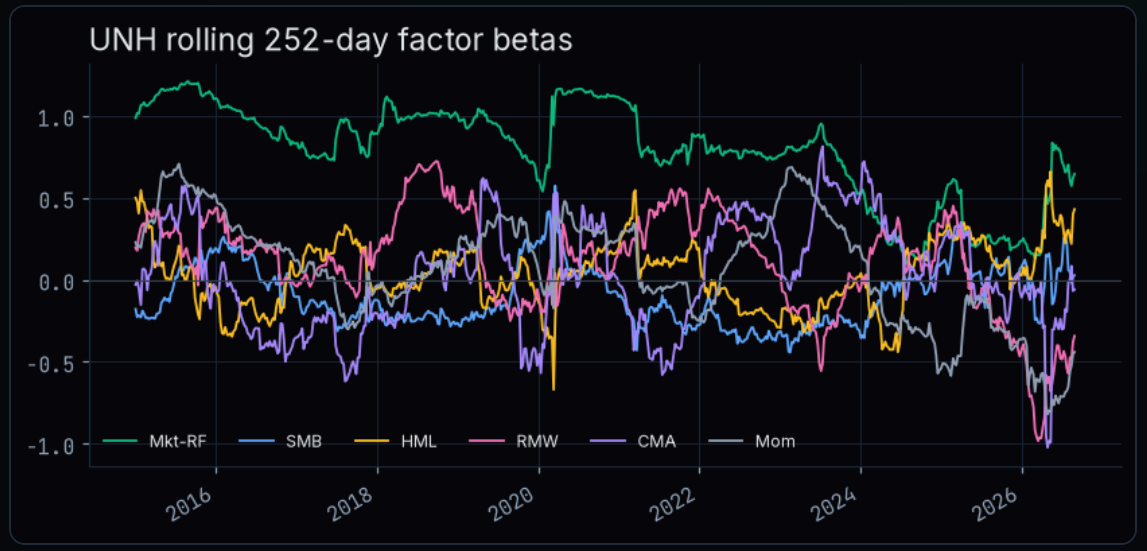


Base FCF $16.1B (FY2025 operating cash flow - capex (SEC 10-K)). Stage-1 growth 13.7% fades to 3.0% terminal growth over 10 years. Value $526.64. The price implies 6.0% growth.

EXPOSURES

Factor exposures: what you are really buying

Factor	Beta	t-stat
Mkt-RF	+0.82	13.3
SMB	-0.12	-2.0
HML	+0.13	1.8
RMW	+0.26	4.3
CMA	+0.09	0.8
Mom	+0.08	1.7
Alpha (annual)	5.9%	0.8



FF5 plus momentum, daily, 2014-01-03 to 2026-08-31. R^2 0.23: most of UNH's risk is stock-specific. After the market, the profitability (RMW) tilt has the strongest t-stat. Alpha is not significant.

RISK

Risk profile

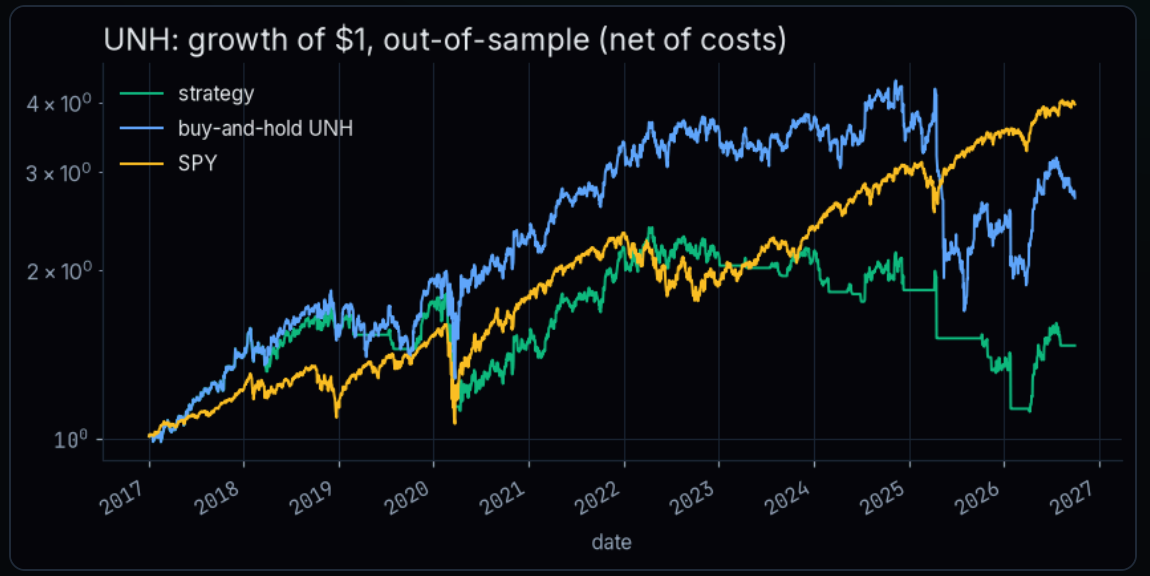
	Ann. vol	Beta vs SPY	Max drawdown	1-day 95% VaR	1-day 95% ES
Full sample: UNH	28.6%	0.78	-61.4%	2.3%	4.0%
Full sample: SPY	17.1%	1.00	-33.7%	1.6%	2.6%
Trailing 3y: UNH	36.8%	0.25	-61.4%	2.8%	5.8%
Trailing 3y: SPY	15.3%	1.00	-18.8%	1.4%	2.1%

UNH is more volatile than SPY and has suffered a -61.4% drawdown. Its low recent beta (0.25 trailing 3y) reflects stock-specific moves, not safety.

EVIDENCE

Backtested signal: it does not help

	CAGR	Sharpe (rf = 0)	Max DD
Trend signal	4.0%	0.29	-53.3%
Buy-and-hold UNH	10.8%	0.49	-61.4%
SPY	15.2%	0.87	-33.7%



Out-of-sample `2017-01-03` to `2026-09-30`, walk-forward window choice, next-day execution, `10` bp per switch. Both the signal and buy-and-hold UNH trail SPY on Sharpe. This is a valuation call, not a momentum trade.

SCENARIOS

Bull, base and bear

	Stage-1 growth	WACC	Value	vs price
Bear	7.6%	8.5%	$354.47	-3.4%
Base	13.7%	8.0%	$526.64	+43.5%
Bull	19.4%	7.5%	$764.68	+108.3%



The bear case is -3.4%, below today's price. Growth is set at the lower quartile, median and upper quartile of UNH's own rolling 5-year FCF growth, and WACC moves by ±0.5%. No probability weights are applied, because there is no data-based way to set them. All three cases still use the 5y beta that sets WACC at 8.0%. With the full-sample beta, the base case alone falls to -12.5% (slide 11).

CATALYSTS

What to monitor

- **Next 10-K FCF.** FY2025 FCF was **$16.1B**. The price implies **6.0%** a year from here. A further decline would break the thesis.
- **Operating margin.** It is **4.2%** against **8.7%** in FY2023. A recovery is the upside driver.
- **Beta re-estimate.** If the 5y beta drifts toward the full-sample **0.78**, WACC rises to **10.5%**.

No dated catalysts are listed: this project uses only the frozen data and does not forecast events.

RISKS

Key risks and what would change my mind

- **The bear case loses money.** The bear value of `$354.47` is `-3.4%` vs the price.
- **The discount rate rests on a low beta.** The 5y beta of `0.41` gives the `8.0%` WACC. The full-sample beta of `0.78` gives a `10.5%` WACC and a value of `$321.29` (`-12.5%`). *On that beta the long case disappears.*
- **FCF is falling.** It fell from `$25.7B` (FY2023) to `$16.1B` (FY2025), while stage-1 growth uses the `13.7%` long-run median.
- **The track record is weak.** Buy-and-hold UNH's Sharpe of `0.49` trails SPY's `0.87`, with a `-61.4%` max drawdown.

> **I would drop the long** if the next 10-K shows FCF still falling, or if a re-estimated beta pushes WACC to the point where the base value falls below the price.

METHOD

Methodology and reproducibility

- As-of `2026-09-30`. Data is frozen with SHA-256 hashes: Yahoo Finance prices and share counts, SEC EDGAR XBRL 10-K facts (filed on or before the as-of date), the Ken French library (factors end `2026-08-31`), and the FRED 10-year yield.
- Notebooks `02` to `06` produce every figure. `results/summary.json` holds every number. A test checks each number on these slides against that file.
- `make all` rebuilds everything offline and runs the tests.

APPENDIX

Appendix A: rejected candidates

	Price	DCF value	vs price	Bear	Bull	Price-implied growth
COST	$910.34	$410.96	-54.9%	$284.63	$496.94	38.5%

V (first pick, archived). Dropped after M3: the base DCF was well below the price, even the bull case showed a loss, and the price implied far more FCF growth than V's history. See `results/archive/V_summary.json`.

COST (third on the shortlist). Every scenario, including bull at $496.94, is below the price. The price implies 38.5% stage-1 growth.

All three were run with identical code and assumptions. Nothing was tuned to make a long case appear.

APPENDIX

Appendix B: assumptions

terminal growth	Assumption: roughly long-run nominal US GDP growth; kept below the 10y yield
fade years	Assumption: growth fades linearly from stage-1 to terminal over 10 years
scenario wacc shift	Assumption: +/-0.5pp WACC in bear/bull
trading cost bps	Assumption: 10 bp per one-way switch (commission plus spread for a mega-cap)
sma grid	Assumption: candidate trend-filter windows (trading days)
train years	Assumption: walk-forward training window
cash return	Assumption: cash earns 0 in the backtest (conservative for the strategy)
beta window years	Convention: 5y of daily returns for the CAPM beta